A PILLOWCASE ACTION COMPLEX $C(K)$ AND A FILTERED MAPPING CONE FOR THE CHERN–SIMONS FILTERED INSTANTON HOMOLOGY OF $1/n$ SURGERIES

Abstract. Let K be a knot in an integer homology sphere Y whose Alexander polynomial has no roots on the unit circle and whose irreducible SU(2) representation variety is a smooth compact 1-manifold. We define a complex $C(K)$ from the image Γ_K of this variety in the pillowcase, together with a Chern–Simons action, a grading, and the two instanton complexes $CI(Y)$ and $CI(Y_0(K), w)$ as an A_∞-module over the two-line algebra of the pillowcase. We give a mapping cone formula for the Chern–Simons filtered irreducible instanton complex of $Y_{1/n}(K)$: $CI(Y_{1/n}) \cong \mathrm{Cone}\big(CI(Y_{1/(n-1)}) \to CI(Y_0, w)\big)$ as complexes, with the Chern–Simons value of each generator given exactly by a Kirk–Klassen integral along Γ_K plus a signed area $\delta_n(x)$ in the pillowcase. The constant-shift version (which is what the level bounds of Daemi–Lidman–Miller Eismeier control) is shown to be non-exact by the figure-eight knot, and is used only as a check. The exact statement is obtained from Floer's own proof of the triangle via collar holonomy perturbations, matched with Lagrangian surgery in the pillowcase. What is proved, what is numerically verified (trefoil, figure-eight), and what remains conjectural are separated, and a workflow is specified.

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1. Setting and standing hypotheses

Y is an integer homology sphere, $K \subset Y$ a knot, $X = Y \setminus \nu K$, with meridian μ and Seifert longitude λ on $\partial X = T^2$. $R(X) = \mathrm{Hom}(\pi_1 X, \mathrm{SU}(2))/\mathrm{conj}$, and $R^{\mathrm{irr}}(X)$ is its irreducible part (an open subset). For $\rho \in R(X)$ choose a conjugate with $\rho(\mu) = \mathrm{diag}(e^{2\pi i a}, e^{-2\pi i a})$, $\rho(\lambda) = \mathrm{diag}(e^{2\pi i b}, e^{-2\pi i b})$; the pair $(a, b) \in T^2 = \mathbb{R}^2/\mathbb{Z}^2$ is defined up to $(a, b) \mapsto (-a, -b)$, and

$$r\colon R(X) \to P := T^2/\pm, \qquad \rho \mapsto [(a, b)],$$

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is the restriction to the boundary. P is the pillowcase, a sphere with four orbifold points (the corners), the images of the central representations $\rho(\mu), \rho(\lambda) \in \{\pm 1\}$. We write $\Gamma_K := r(R^{\mathrm{irr}}(X))$.

Irreducible instanton homology: for an integer homology sphere Z, $CI_*(Z)$ denotes Floer's complex generated by the irreducible flat connections (after a small holonomy perturbation), $\mathbb{Z}/8$-graded, with its Chern–Simons filtration. We work over $\mathbb{F} = \mathbb{Z}/2$ unless stated otherwise; orientations are a separate issue (§8).

1.1. **Hypotheses.**

(H1) $\Delta_K(e^{2i\theta}) \neq 0$ for all θ.

(H2) $R^{\mathrm{irr}}(X)$ is a smooth 1-manifold, and $H^1(X; \operatorname{ad}\rho)$ is 1-dimensional for every $\rho \in R^{\mathrm{irr}}(X)$ (the Zariski tangent space equals the tangent space).

For $Y \neq S^3$ we add

(H3) every irreducible flat connection ρ on Y is nondegenerate and has $\rho(K) \neq \pm 1$.

1.2. **Consequences.**

Proposition 1.1. *Assume (H1)–(H2).*

- *(a) $R^{\mathrm{irr}}(X)$ is compact: a finite disjoint union of circles $\gamma_1, \dots, \gamma_m$. Its image Γ_K is a finite union of immersed closed curves in P: r is an immersion.*
- *(b) If $Y = S^3$, Γ_K avoids the two edges $a \in \{0, \frac{1}{2}\}$ of P, in particular the corners. In general, Γ_K meets the corners only at representations with $\rho(\mu) = \pm 1$; those with $\rho(\mu) = 1$ are the irreducible flat connections on Y.*
- *(c) $R(X)$ is invariant under $\chi\colon \rho \mapsto \rho \cdot \chi$, where $\chi\colon \pi_1 X \to \{\pm 1\}$ is the nontrivial character; on T^2 this is $(a, b) \mapsto (a + \frac{1}{2}, b)$. Hence Γ_K is invariant under the translation $a \mapsto a + \frac{1}{2}$.*

Proof. (a) By Klassen's theorem (and its extension to homology spheres), an abelian representation with $\rho(\mu) = e^{i\theta}$ is a limit of irreducibles only if $\Delta_K(e^{2i\theta}) = 0$. Under (H1) no abelian representation is such a limit, so $R^{\mathrm{irr}}(X)$ is closed in the compact space $R(X)$. With (H2) it is a compact 1-manifold. The image of $dr\colon H^1(X; \operatorname{ad}\rho) \to H^1(T^2; \operatorname{ad}\rho)$ is Lagrangian for the cup-product symplectic form, hence 1-dimensional; with $\dim H^1(X; \operatorname{ad}\rho) = 1$, dr is injective. (b) If $Y = S^3$ then μ normally generates $\pi_1 X$, so $\rho(\mu) = \pm 1$ forces ρ central. In general $\rho(\mu) = 1$ means ρ factors through $\pi_1 Y$. (c) λ lies in the commutator subgroup, so $\chi(\lambda) = 1$. □

1.3. **Surgery lines.**

For a slope p/q (with p, q coprime, $p/q = 1/0$ for Y itself) put

$$L_{p/q} := \{pa + qb \equiv 0\}, \qquad L^w_{p/q} := \{pa + qb \equiv \tfrac{1}{2}\} \subset T^2,$$

and their images in P. Flat SU(2) connections on $Y_{p/q}(K)$ restrict to representations of $\pi_1 X$ with $\rho(\mu^p \lambda^q) = 1$, i.e. to points of $L_{p/q}$. For p even, $L^w_{p/q}$ describes the flat connections on the nontrivial SO(3) bundle; in particular $L^w_0 = \{b = \frac{1}{2}\}$ describes $I(Y_0(K), w)$ with w the nontrivial bundle (admissible since Y_0 has $b_1 = 1$).

Proposition 1.2 (generators)**.** *Assume (H1)–(H2), and $p \neq 0$.*

- *(a) The irreducible flat connections on $Y_{p/q}(K)$ are in bijection with $r^{-1}(L_{p/q}) \subset R^{\mathrm{irr}}(X)$. The only reducible flat connection on $Y_{1/n}(K)$ is the trivial one.*
- *(b) Such a flat connection ρ is nondegenerate ($H^1(Y_{p/q}; \operatorname{ad}\rho) = 0$) if and only if Γ_K is transverse to $L_{p/q}$ at $r(\rho)$.*
- *(c) The flat connections on $(Y_0(K), w)$ are in bijection with $r^{-1}(L^w_0)$, all irreducible; nondegeneracy is as in (b).*

Proof. (a) A representation of $\pi_1 X$ with $\rho(\mu^p\lambda^q) = 1$ extends uniquely over the solid torus, and $\pi_1 X \to \pi_1 Y_{p/q}$ is onto, so irreducibility is preserved both ways. An abelian ρ has $\rho(\lambda) = 1$, so $\rho(\mu)^p = 1$; for $p = \pm 1$ this is trivial. (b) Mayer–Vietoris for $\operatorname{ad}\rho$ along T^2: since $\rho|_{T^2}$ is non-central abelian (by Proposition 1.1(b) and (H3)), $H^0(T^2) = \mathbb{R} = H^0(\text{solid torus})$ and $H^1(T^2; \operatorname{ad}\rho) = T_{(a,b)}T^2$ is 2-dimensional. Then $H^1(Y_{p/q}; \operatorname{ad}\rho) = \ker\big(H^1(X) \oplus H^1(S^1 \times D^2) \to H^1(T^2)\big)$, with $H^1(X) = T_\rho R^{\mathrm{irr}}(X)$ mapping injectively by dr and $H^1(S^1 \times D^2) = \mathbb{R}$ mapping onto the tangent line of $L_{p/q}$. The kernel vanishes iff the two lines are transverse. (c) is the same with $\rho(\lambda) = -1$; abelian representations have $\rho(\lambda) = 1$, so none occur. □

Remark 1.3. Transversality can be achieved for every slope by an arbitrarily small holonomy perturbation supported in a collar of T^2; in the pillowcase this replaces $L_{p/q}$ by a C^1-close curve $L'_{p/q}$ (Herald's collar perturbations). The regularity of instanton moduli spaces may require further small perturbations. All statements below are stable under such perturbations: the generator data move by $O(\epsilon)$ and the filtered homotopy type is taken in the limit, as in the definition of Daemi's Γ_Y.

2. Chern–Simons as symplectic action in the pillowcase

Normalise $\mathrm{cs}\colon \mathcal{B}(Z) \to \mathbb{R}/\mathbb{Z}$ so that the energy of an instanton on $\mathbb{R} \times Z$ from α to β is $\mathrm{cs}(\alpha) - \mathrm{cs}(\beta)$ (lifted to $\mathbb{R}$), and so that the trivial connection has $\mathrm{cs} = 0$.

2.1. **Forms and primitives.** On $\mathbb{R}^2 \ni (a, b)$ let $\omega = 2\,da \wedge db$ and $\lambda = 2b\,da$. The factor 2 makes ω descend to P with total area 1, matching the $\mathbb{Z}$-periodicity of cs. For an oriented curve $C \subset \mathbb{R}^2$ with a base point c_0, set $f_C(x) := -\int_{c_0}^{x} \lambda$ along C.

Lemma 2.1 (change of basis)**.** *For* $\left(\begin{smallmatrix} p & q \\ r & s \end{smallmatrix}\right) \in SL(2,\mathbb{Z})$ *and* $a' = pa + qb$, $b' = ra + sb$: $2b'\,da' - 2b\,da = dg$ *with* $g = 2qr\,ab + pr\,a^2 + qs\,b^2$. *On* $L_{p/q} = \{a' = 0\}$ *parametrised by* $(a, b) = s(q, -p)$ *one has* $g = pq\,s^2$.

Proof. Expand $b'\,da' = (ra + sb)(p\,da + q\,db)$ and use $ps - qr = 1$; the terms regroup as $b\,da + d(qr\,ab + \frac{1}{2}pr\,a^2 + \frac{1}{2}qs\,b^2)$. On the line, substitute. □

2.2. **The Kirk–Klassen formula.** Kirk–Klassen [1]: if ρ_t is a path in $R(X)$ with boundary angles $(a(t), b(t))$ lifted continuously to $\mathbb{R}^2$, and ρ_0, ρ_1 extend over the same Dehn filling Z, then, in a basis of $H_1(T^2)$ in which the filling curve is the first basis element,

$$\mathrm{cs}(\rho_1) - \mathrm{cs}(\rho_0) \equiv -\int_0^1 2b\,\dot{a}\,dt \pmod{\mathbb{Z}}.$$

Combining with Lemma 2.1:

Proposition 2.2 (Chern–Simons is the action)**.** *Fix* (r, s) *with* $ps - qr = 1$ *and put* $Q(a, b) := pr\,a^2 + 2qr\,ab + qs\,b^2$ *(the* g *of Lemma 2.1). Let* $x, y \in r^{-1}(L_{p/q})$ *lie on one component* γ *of* $R^{\mathrm{irr}}(X)$ *and lift a path in* γ *from* x *to* y *to* $\tilde{\gamma} \subset \mathbb{R}^2$ *with endpoints* $\tilde{x}, \tilde{y}$. *Then*

$$\mathrm{cs}(y) - \mathrm{cs}(x) \equiv \big[f_{\tilde{\gamma}}(\tilde{y}) - Q(\tilde{y})\big] - \big[f_{\tilde{\gamma}}(\tilde{x}) - Q(\tilde{x})\big] \pmod{\mathbb{Z}}.$$

Changing (r, s) *changes* Q *by* $(pa + qb)^2 \in \mathbb{Z}$ *at the lifted points, so the right side is well defined. On* $L_{p/q}$ *itself,* $Q = pq\,s^2$ *in the parametrisation* $(a, b) = s(q, -p)$*; but the lifted endpoints lie on translates* $pa + qb = k$, *and* Q *must be evaluated there, not on the projection to* $L_{p/q}$. *For* L_0^w *take* $(p, q) = (0, 1)$*:* $Q = 2r\,ab + s\,b^2$ *with* $s = 1$, *defined modulo* $b^2 = \frac{1}{4}$ *on* $b \in \frac{1}{2} + \mathbb{Z}$.

Proof. Kirk–Klassen in the adapted basis $(a', b') = (pa + qb,\ ra + sb)$, in which the filling curve is the first basis element and the formula is $-\int 2b'\,da'$; rewrite by Lemma 2.1. This is the form given by Andersen–Hansen [21, eq. (5.67)], who correct a sign in [1]. □

The right-hand side is the symplectic action $A(x) := f_{\tilde\gamma}(\tilde x) - Q(\tilde x)$ of the intersection point. *Numerically verified on all* 26 *flat connections of the six homology-sphere surgeries* $\pm1, \pm1/2, \pm1/3$ *on the trefoil, and on* ±1 *surgeries on the figure-eight* (§7).

Remark 2.3. For $p/q = 1/n$ the formula reads $\mathrm{cs}(y) - \mathrm{cs}(x) \equiv -\int 2b\,da + n\,(b(y)^2 - b(x)^2)$, which is the form used by Nozaki–Sato–Taniguchi [12, Thm. 7.1] (with their sign convention for the integral). An integer-valued lift of the Kirk–Klassen formula, which is what the filtered complex needs, is Boden–Herald–Kirk–Klassen [18, Thm. 5.5]: $\mathrm{cs}(A_1) = -c + 2\int n\,\dot m\,dt$ with c a linking number of an explicit loop.

So *the Chern–Simons filtration on the generators of* $CI(Y_{p/q})$ *coming from one component of* $R^{\mathrm{irr}}(X)$ *is the action filtration of the pillowcase Floer complex.*

Lemma 2.4 (periods)**.** *f_γ does not in general descend to γ. By* [2]*, cs_X is a section over $R(X)$ of the circle bundle $E(\partial X) = \big(\mathbb{R}^2 \times U(1)\big)/\!\approx$ with $(a,b;z) \approx (a+1,b;ze^{-8\pi ib}) \approx (a,b+1;ze^{8\pi ia}) \approx (-a,-b;z)$, and $e^{2\pi i f_\gamma}$ is the pull-back of that section to the lift. Precisely: if one circuit of γ carries the lift $(a,b)(0)$ to $\varepsilon(a,b)(0) + (m,k)$, $\varepsilon = \pm1$, then*

$$\oint_\gamma 2b\,da \equiv 2\varepsilon k\,a(0) \pmod{\mathbb{Z}},$$

so the period is an integer iff $k = 0$, and for $K \subset S^3$ (where $\varepsilon = 1$, $m = 0$) the lift-independent statement is $\oint_\gamma 2a\,db \in \mathbb{Z}$.

Proof. Along normal-form paths $d\,\mathrm{cs}_X = \pm\lambda$; around the loop the endpoint connections differ by a gauge transformation whose restriction to ∂X is $\exp(2\pi(mx+ky)\sigma)$, not boundary-trivial, contributing $\pm(mb - ka)$ modulo the integer bulk degree; the Weyl element is constant and contributes nothing. □

For the figure-eight knot, $k = 2$, $\oint 2b\,da = 4a(0) - 1$ (numerically, to 10^{-6}, at every base point) and $\oint 2a\,db = 1$ (§7). The first draft of this lemma claimed $\oint 2b\,da \in \frac12\mathbb{Z}$; that is false.

2.3. **Absolute constants: the first correction terms.** Proposition 2.2 determines cs on the generators coming from one component γ_i only up to a constant $c_i \in \mathbb{R}/\mathbb{Z}$: there is no path in $R(X)$ from the trivial connection to γ_i, precisely because of (H1). These constants are genuine knot invariants (they are the Chern–Simons invariants of flat connections on fillings), and they are computable:

(i) through the $SL(2,\mathbb{C})$ character variety, where the Kirk–Klassen formula holds with complex (a,b) [2] and the irreducible curve containing γ_i reaches the abelian locus at roots of Δ_K off the unit circle; the imaginary part of the integral must vanish at the end, which is a check;
(ii) or from one known filling (e.g. a Brieskorn sphere), which calibrates c_i for every slope at once.

Thus the Chern–Simons data of K is: the curve Γ_K, the primitives f_{γ_i}, and the constants c_i.

3. The complex $C(K)$

3.1. **Gradings.** Floer's $\mathbb{Z}/8$ grading of a flat connection on $Y_{p/q}$ is a spectral flow. Kirk–Klassen–Ruberman [3] split it along T^2: for $x, y \in r^{-1}(L_{p/q})$ on one component γ,

$$\mathrm{gr}(y) - \mathrm{gr}(x) \equiv \mathrm{Mas}\big(\tilde\gamma|_{x\to y},\, L\big) \pmod 8,$$

where Mas is the Maslov index of the path of tangent lines of $\tilde\gamma$ (an immersed curve, by Proposition 1.1) against the fixed line L in $(\mathbb{R}^2,\omega)$; the exact form of the splitting formula, including

the contribution of the solid torus, is Task 4 of §8. Within a component this is computable from the picture. Across components there are constants $d_i \in \mathbb{Z}/8$, again fixed by one known filling. The $\mathbb{Z}/2$ reduction is Herald's sign [5], and $\sum_x (-1)^{\mathrm{gr}(x)}$ over $\Gamma_K \cap L_{1/n}$ is the Casson invariant of $Y_{1/n}$ (Herald's formula).

3.2. **Two instanton complexes and the two-line algebra.** Let $A := CI(Y)$ and $B := CI(Y_0(K), w)$, both $\mathbb{Z}/8$-graded, with the Chern–Simons filtrations of §2 (generators $r^{-1}(L_{1/0})$ and $r^{-1}(L_0^w)$). For $Y = S^3$, $A = 0$.

In the pillowcase, $L_{1/0} = \{a = 0\}$ and $L_0^w = \{b = \frac{1}{2}\}$ meet in exactly one point of T^2, the corner $c = (0, \frac{1}{2})$. Their endomorphism algebra in the Fukaya category of T^2 is the path algebra

$$\mathcal{A} = \mathbb{F}\langle e_A, e_B,\ \mathsf{p}\colon A \to B,\ \mathsf{q}\colon B \to A,\ \varepsilon_A, \varepsilon_B\rangle / (\mathsf{qp} = \varepsilon_A,\ \mathsf{pq} = \varepsilon_B,\ \varepsilon^2 = 0,\ \varepsilon\mathsf{p} = \mathsf{p}\varepsilon,\ \ldots),$$

p, q the corner point in the two directions, ε the degree-one generator of $H^*(S^1) = \mathrm{Hom}(L, L)$. This is the instanton counterpart of the bordered torus algebra.

Definition 3.1 (the complex $C(K)$)**.** $C(K)$ is the filtered, $\mathbb{Z}/8$-graded A_∞-module over $\mathcal{A}$ with underlying filtered complex $A \oplus B$ and structure maps

$$\begin{aligned}
m_2(-, \mathsf{p}) &= F_W\colon A \to B, && \text{the 2-handle cobordism } W\colon Y \to Y_0 \text{ (bundle } w),\\
m_2(-, \mathsf{q}) &= F_{W'}\colon B \to A, && \text{the reverse cobordism } Y_0 \to Y,\\
m_2(-, \varepsilon_A) &= u_A, \quad m_2(-, \varepsilon_B) = u_B, && \text{the point-class (degree } -4) \text{ operators,}
\end{aligned}$$

and higher maps m_k given by counts on the composite cobordisms with the corresponding handle decompositions, e.g. $m_3(-, \mathsf{p}, \mathsf{q})$ the homotopy $F_{W'}F_W \simeq u_A$ on the composite $Y \to Y_0 \to Y$. Together with the generator data $(\Gamma_K, f_{\gamma_i}, c_i, \mathrm{gr}, d_i)$ of §2.

Status. The underlying complexes, $m_2(-, \mathsf{p})$, $m_2(-, \mathsf{q})$, u_A, u_B and the homotopy $m_3(-, \mathsf{p}, \mathsf{q})$ exist rigorously: they are the maps and homotopies of the surgery exact triangle (Floer; Braam–Donaldson [6]; Kronheimer–Mrowka's triangle detection [8]), and the energy of an instanton on a cobordism is nonnegative, so each is filtered after an explicit shift (§4). The full A_∞-structure is the content of a bordered/Atiyah–Floer theory for the torus boundary, which is conjectural. For $1/n$ surgeries only the listed maps are needed (Theorem 4.2).

Conjecture 3.2 (pillowcase model)**.** *Under (H1)–(H3), $C(K)$ is A_∞-quasi-isomorphic, as a filtered module, to the Fukaya module* $\mathrm{Hom}(\tilde{\Gamma}_K,\ L_{1/0} \oplus L_0^w)$ *in the $\mathbb{Z}/2$-equivariant Fukaya category of T^2, with $\tilde{\Gamma}_K$ equipped with a bounding cochain supported at its double points, over the Novikov field (the products μ^2 on T^2 are infinite sums over triangles), and the action/grading data of §2–3. Differentials are counts of immersed bigons in T^2 avoiding the corners.*

Evidence: generators, nondegeneracy, Chern–Simons values and gradings match (Propositions 1.2, 2.2, and [3, 5]); the Euler characteristic matches by Herald's formula; the exact triangle matches the Dehn-twist triangle (§4). What is missing is the identification of instantons on $\mathbb{R} \times Y_{1/n}$ with bigons, the analogue of the Atiyah–Floer conjecture for torus boundary.

4. The filtered mapping cone

4.1. **The corner offsets.** For $n \geq 1$ let $p_n \in T^2$ be the intersection point of $L_{1/(n-1)} = \{a + (n-1)b = 0\}$ (for $n = 1$: $L_{1/0}$) with $L_0^w = \{b = \frac{1}{2}\}$; it is the corner $(0, \frac{1}{2})$ for n odd and $(\frac{1}{2}, \frac{1}{2})$ for n even.

Lemma 4.1 (corner action)**.** *With the standard lifts (origin-based primitive on $L_{1/(n-1)}$, $(0,\frac{1}{2})$-based on L_0^w) the action $A(p_n) = f_{L_{1/(n-1)}}(\tilde{p}_n) - f_{L_0^w}(\tilde{p}_n)$ equals $\frac{n-1}{4}$ modulo $\mathbb{Z}$; but the primitive on the w-line is itself only defined modulo $\frac{1}{4}$ (Proposition 2.2), so the invariant content is*

$$A(p_n) \in \tfrac{1}{4}\mathbb{Z}.$$

Proof. At $b = \frac{1}{2}$: $s = -\frac{1}{2}$, $a = -(n-1)/2$, $f_{L_{1/(n-1)}} = -(n-1)/4$, $f_{L_0^w} = (n-1)/2$, difference $\equiv (n-1)/4$. Lifts along $L_{1/(n-1)}$ change f by $-(n-1)(k^2-k) \in 2\mathbb{Z}$; the choice of d in Q on the w-line changes it by $b^2 = \frac{1}{4}$. □

So the pillowcase reproduces the quarter-integrality of instanton energies on the 2-handle cobordism W_n (whose bundle has $w^2 \neq 0$), and nothing finer. The actual filtration shift of F_n is its *level* in the sense of Daemi–Lidman–Miller Eismeier [17]: $L_n = -c_n^2/4 - \eta(W_n, c_n)$, with $\eta \geq 0$ the minimal energy defect; for the ± 1 triangle they prove $L = 0$ for the Floer map and $L \leq \frac{1}{4}$ for the distance-two maps.

4.2. **The Dehn-twist triangle.** The (inverse) Dehn twist along $L_0^w = \{b = \frac{1}{2}\}$ carries $L_{1/(n-1)}$, up to Hamiltonian isotopy, to the line $L_{1/n} = \{a + nb = 0\}$ through the origin, *not* to $L_{1/n}^w$ (Lemma 5.1; the linear shear fixing L_0^w pointwise lands on $L_{1/n}^w$ but is not Hamiltonian isotopic to the twist). Seidel's exact triangle [15], in the Fukaya category of T^2 over the Novikov field, reads

$$L_{1/(n-1)} \xrightarrow{\cdot p_n} L_0^w \longrightarrow L_{1/n} \longrightarrow L_{1/(n-1)}[1],$$

and applying $\mathrm{Hom}(\tilde{\Gamma_K}, -)$ gives $\mathrm{Hom}(\tilde{\Gamma_K}, L_{1/n}) \simeq \mathrm{Cone}\big(\mathrm{Hom}(\tilde{\Gamma_K}, L_{1/(n-1)}) \xrightarrow{m_2(-,p_n)} \mathrm{Hom}(\tilde{\Gamma_K}, L_0^w)\big)$. This is the pillowcase shadow of Floer's exact triangle for the triple $(Y_{1/(n-1)},\ (Y_0, w),\ Y_{1/n})$, with the bundle w on Y_0 only, as it should be. (A test object: the line $L_{1/n}^w$ is concurrent with the two others at p_n, and $\mu^2(p_n, -)$ is an isomorphism on it, so it cannot be the cone.)

4.3. **Relation to Daemi–Lidman–Miller Eismeier.** A filtered version of Floer's triangle for ± 1 surgery already exists: Daemi–Lidman–Miller Eismeier [17] prove that a cobordism map $(W, c)_*$ has degree $-2c^2$ and level $-c^2/4 - \eta(W, c)$, that the Floer triangle map has degree 0 and level 0, and that in the distance-two triangle the maps have level $\leq \frac{1}{4}$. The offset $\frac{n-1}{4}$ below is the same quantity seen from the pillowcase: the level of a cobordism map is the action of the corresponding corner morphism. What is added here is (i) the generator-level description of all three complexes and their filtrations in terms of one curve Γ_K, and (ii) the iterated form for $1/n$.

4.4. **The theorem.**

Theorem 4.2 (filtered mapping cone for $1/n$ surgery)**.** *Assume (H1)–(H3), $n \geq 1$, and that $L_{1/(n-1)}$ and $L_{1/n}$ are transverse to Γ_K (after collar perturbation). Let $F_n\colon CI(Y_{1/(n-1)}) \to CI(Y_0, w)$ be the map of the 2-handle cobordism W_n.*

- *(a)* (Floer, Braam–Donaldson) *$CI(Y_{1/n}) \simeq \mathrm{Cone}(F_n)$ as $\mathbb{Z}/8$-graded complexes, the grading on* Cone *being $\mathrm{gr}_{CI(Y_{1/(n-1)})} + \delta_n$ on the first summand and $\mathrm{gr}_{CI(Y_0,w)}$ on the second, for a constant $\delta_n \in \mathbb{Z}/8$ determined by the index of W_n (computed in §7).*
- *(b)* (filtration) *Give* Cone(F_n) *the filtration* cs *on $CI(Y_0, w)$ and* cs $+ L_n$ *on $CI(Y_{1/(n-1)})$, where $L_n \in \frac{1}{4}\mathbb{Z} - \eta_n$ is the level of F_n. Then the triangle-detection maps* Cone$(F_n) \rightleftarrows CI(Y_{1/n})$ *are filtered after shifts by the levels of the other two cobordism maps, and are homotopy inverses through filtered homotopies. Hence the Chern–Simons filtered homotopy type of $CI(Y_{1/n})$ is determined, up to these explicit shifts, by $CI(Y_{1/(n-1)})$, $CI(Y_0, w)$ and F_n; in particular Daemi's $\Gamma_{Y_{1/n}}$, the Nozaki–Sato–Taniguchi r_s and the*

ℓ-invariant of [17] *are sandwiched between the corresponding invariants of the cone with the two shifts.*

(c) (generator level) *The generators of each of the three complexes, with their* cs *values and gradings, are the intersection data of §2–3. The cone does* not *compute the* cs *values of the flat connections on $Y_{1/n}$ as a multiset: filtered homotopy equivalence preserves the filtered homotopy type, not generator levels, and the trefoil example shows the multisets differ (§7).*

Proof sketch. (a) is the surgery exact triangle for the three fillings of slope $1/(n-1)$, 0 (bundle w) and $1/n$, which are pairwise at distance one. (b) The maps of the triangle detection lemma (Kronheimer–Mrowka [8, §7]) count instantons on W_n, on the second cobordism $W_n'\colon Y_0 \to Y_{1/n}$, and on the composites with up to four handles; each shifts cs by the energy, which is nonnegative, plus a topological constant fixed by the bundle on the cobordism. So every map is filtered once the summands are shifted by these constants, and the constants for the two inclusions and projections of the cone are forced to be equal by the homotopy identities. The constants are the levels of [17, Prop. 3.4]; the pillowcase corner action (Lemma 4.1) accounts for their quarter-integral part. (c) is Propositions 1.2 and 2.2. □

What is not computed here: the levels L_n for $n \geq 2$ (a topological computation on W_n in the manner of [17]), and the differential of $CI(Y_0, w)$ itself (an instanton count, Conjecture 3.2).

4.5. Iteration, $Y = S^3$, negative slopes, slope 2.

Corollary 4.3. *For $Y = S^3$ (so $CI(Y) = 0$): $CI(S^3_1(K)) \simeq CI(S^3_0(K), w)$ with a grading shift and no filtration shift, and for $n \geq 2$ the filtered complex $CI(S^3_{1/n}(K))$ is filtered homotopy equivalent to a twisted complex*

$$\Big(\bigoplus_{j=1}^{n} B[\delta^{(j)}]\{L^{(j)}\},\quad D = \partial_B + \sum_{i<j} D_{ij}\Big), \qquad B = CI(S^3_0(K), w),$$

where $D_{j,j+1}$ is the Dehn-twist endomorphism $T_j = F_{j+1} \circ$ (the equivalence $B \simeq CI(S^3_{1/j})$) and the D_{ij}, $j > i+1$, are the higher homotopies of the iterated cones.

Negative slopes $-1/n$ follow by applying the theorem to the mirror. For slope ± 2 the triple $(1/0,\ 1/1,\ 2/1)$ is at pairwise distance one, but Y_2 has a nontrivial central flat connection (the corner $(\frac{1}{2}, 0)$ lies on L_2), so the irreducible-only complex must be replaced by the SO(3)-equivariant one at that corner; the pillowcase picture predicts a single extra generator with $\mathrm{cs} = 0$ and a corner bigon correction. We leave ± 2 as a stated problem.

5. The exact statement: Lagrangian surgery and collar perturbations

Theorem 4.2(b) is an inequality-type statement (as in [17]), and no statement of that shape can be exact: for the figure-eight knot and $n = 1$ the cone is $CI(S^3_0(4_1), w)$, whose two generators have cs differing by $\frac{1}{2}$ (modulo the $\frac{1}{4}$ ambiguity of the w-line), while the two generators of $CI(\pm\Sigma(2,3,7))$ differ by $\frac{4}{7}$. The exact relation has generator-dependent corrections, and they are areas.

5.1. The zigzag.

Let $Z_n \subset T^2$ be the Lagrangian surgery of $L_{1/(n-1)}$ and L_0^w at the corner p_n: the two lines with the crossing resolved, smoothed in an ϵ-neighbourhood of p_n. It is a closed curve in the class of $L_{1/n}$.

Lemma 5.1 (zero flux)**.** *Z_n is Hamiltonian isotopic in T^2 to the line $L_{1/n} = \{a + nb = 0\}$ through the origin (not to $L^w_{1/n}$).*

Proof. In $\mathbb{R}^2$ one circuit of Z_n lifts to the path $\tilde{p}_n \to \tilde{p}_n + v_1 \to \tilde{p}_n + v_1 + v_2$ with $v_1 = (n-1, -1)$, $v_2 = (1, 0)$, and $\tilde{p}_n = (-\frac{n-1}{2}, \frac{1}{2})$. Two isotopic closed curves on T^2 are Hamiltonian isotopic iff the signed area between them vanishes. Against the line $a + nb = c$ the signed area is a linear function of c vanishing exactly when the line passes through the midpoints $\tilde{p}_n + \frac{1}{2}v_1 = (0, 0)$ and $\tilde{p}_n + v_1 + \frac{1}{2}v_2$ of the two legs, i.e. for $c = 0$. □

This is the pillowcase form of the fact that $Y_{1/n}$ carries the trivial bundle.

Proposition 5.2 (exact cone for the zigzag)**.** *As $\epsilon \to 0$, $CF(\tilde{\Gamma_K}, Z_n)$ with its action filtration is isomorphic, as a filtered complex, to*

$$\operatorname{Cone}\big(CF(\tilde{\Gamma_K}, L_{1/(n-1)}) \xrightarrow{\mu^2(-,p_n)} CF(\tilde{\Gamma_K}, L_0^w)\big),$$

the first summand carrying the line action $A(x) = f_\Gamma(\tilde{x}) - (n-1)\tilde{b}^2$ of Proposition 2.2, the second the action $f_\Gamma(\tilde{x}) + 2\tilde{a}\tilde{b} + \frac{n}{4}$ (the Kirk–Klassen primitive on $b \in \frac{1}{2} + \mathbb{Z}$, offset by $\frac{n}{4}$).

Proof. Generators of $CF(\tilde{\Gamma_K}, Z_n)$ are those of the two summands. Bigons between points of the same summand are bigons for that line. A bigon from a point of $L_{1/(n-1)}$ to a point of L_0^w must pass through the surgery region, and for $\epsilon \to 0$ it is a triangle with a corner at p_n: this is $\mu^2(-, p_n)$. The primitive of λ on Z_n restricts to the two line primitives, offset by a constant, up to $O(\epsilon)$. The constant is fixed by requiring both to be primitives of λ on Z_n agreeing at the points where Z_n meets $L_{1/n}$: these are the midpoints $m_1 = \tilde{p}_n + \frac{1}{2}v_1$, $m_2 = \tilde{p}_n + v_1 + \frac{1}{2}v_2$ of the two legs (Lemma 5.1), not the vertex $\tilde{p}_n$, and the mismatch of the two line primitives at m_2 (where $b \in \frac{1}{2} + \mathbb{Z}$) is $n\,b^2 \equiv \frac{n}{4}$. The vertex offset $(n-1)/4$ of Lemma 4.1 is the wrong normalisation for the cone; the numerics confirm $\frac{n}{4}$ on the w-leg and 0 on the other (§7). □

5.2. **From the zigzag to the straight line.** Let ϕ_t be an equivariant Hamiltonian isotopy of T^2 with $\phi_0(Z_n) = Z_n$ and $\phi_1(Z_n) = L_{1/n}$ (Lemma 5.1). Say Γ_K is in *minimal position* for the pair if $\phi_t(Z_n) \cap \tilde{\Gamma_K}$ has constant cardinality; by the class computation this holds iff $|\Gamma_K \cap L_{1/n}| = |\Gamma_K \cap L_{1/(n-1)}| + |\Gamma_K \cap L_0^w|$, which is the case for the figure-eight and all n (counts $2|n| = 2|n-1| + 2$).

Theorem 5.3 (exact filtered cone, pillowcase side)**.** *Assume (H1)–(H3) and minimal position. Then*

- *(a) the isotopy induces a bijection $x \mapsto x'$ from the generators of the cone to $\Gamma_K \cap L_{1/n}$ and an isomorphism of complexes $\operatorname{Cone}(\mu^2(-, p_n)) \cong CF(\tilde{\Gamma_K}, L_{1/n})$;*
- *(b) the action of x' is $A_{\text{cone}}(x) + \delta_n(x)$, where A_{cone} is the cone action of Proposition 5.2 and $\delta_n(x) = \int_{R_n(x)} \omega$ is the signed area of the region in $\mathbb{R}^2$ bounded by the arc of $\tilde{\Gamma_K}$ from $\tilde{x}$ to $\tilde{x}'$, the segment of Z_n from $\tilde{x}$ to the midpoint m of the leg containing $\tilde{x}$, and the segment of $L_{1/n}$ from m to $\tilde{x}'$;*
- *(c) cs on the generators of $CI(Y_{1/n})$ is $A(x') + c_i$, exactly (Proposition 2.2).*

Proof. (a) Between birth–death moments the bigon counts of $CF(\tilde{\Gamma_K}, \phi_t(Z_n))$ are locally constant. (b) is Stokes' theorem for λ along the boundary of $R_n(x)$, the two primitives agreeing at m by construction. (c) is Proposition 2.2. Numerically verified for the figure-eight, $n = 1, 2, 3$, to $2 \cdot 10^{-10}$ (§7). □

Without minimal position, (a) holds after cancelling the birth–death pairs, each of which is an explicit bigon of explicit area. The DLME-type level bounds follow from $|\delta_n(x)| \le$ area swept; the exact statement is (b).

5.3. **The instanton side: Floer's perturbation.** The pillowcase statement has an exact instanton counterpart, and it is Floer's original proof of the triangle [7, 6]. By Sivek–Zentner [20, Thm. 4.5], any area-preserving isotopy of the pillowcase applied to the filling line is approximated by shearing isotopies induced by holonomy perturbations supported in the collar $T^2 \times I \subset Y_{1/n}$. Choose such a perturbation π_n moving $L_{1/n}$ onto (a smoothing of) Z_n.

(i) The critical set of $\mathrm{cs} + \pi_n$ on $Y_{1/n}$ is $r^{-1}(L_{1/(n-1)}) \sqcup r^{-1}(L_0^w) \sqcup \{\theta\}$, all nondegenerate (Proposition 1.2 applied to Z_n).

(ii) The perturbed Chern–Simons values are the cone actions. Precisely (Appendix A): a shear perturbation with Hamiltonian H along $\ell_v = pa + qb$ moves the line by the time-one flow of $H \circ \ell_v$ for $\omega_P = 2\,da \wedge db$, and the perturbed functional on a perturbed-flat collar piece equals the generating function $S = H(\ell) - \frac{1}{2}\ell H'(\ell)$ of the shear, so that $\mathrm{cs}_\pi(\rho_1) - \mathrm{cs}_\pi(\rho_0) \equiv \int_\gamma \lambda_0 - \left[f_{L^\pi}\right]$ with f_{L^π} the primitive of $\lambda_0 = a\,db - b\,da$ along the perturbed line: the difference is twice the signed area between γ and L^π. (The naive guess "add $h(\rho_1) - h(\rho_0)$" is wrong: the curvature in the collar contributes $-\frac{1}{2}\ell H'(\ell)$.)

(iii) Floer's argument gives a two-step filtration of $CI(Y_{1/n}; \pi_n)$ by the two halves of the critical set, with associated graded $CI(Y_{1/(n-1)})$ and $CI(Y_0, w)$ and connecting map F_n; this is an *isomorphism* of complexes $CI(Y_{1/n}; \pi_n) \cong \mathrm{Cone}(F_n)$, not merely a quasi-isomorphism.

(iv) The family $\mathrm{cs} + t\pi_n$, $t \in [0,1]$, gives a continuation map $CI(Y_{1/n}; \pi_n) \to CI(Y_{1/n})$; when the critical points move without birth–death (the instanton form of minimal position), it is an isomorphism of complexes sending each generator x to x' with $\mathrm{cs}(x') = \mathrm{cs}_{\pi_n}(x) + \int_0^1 \left(\partial_t \mathrm{cs}_{t\pi_n}\right)(x_t)\,dt$, the integral being $\delta_n(x)$.

Theorem 5.4 (exact filtered cone, instanton side)**.** *Assume (H1)–(H3), transversality of Γ_K to $L_{1/(n-1)}, L_0^w, L_{1/n}$, and minimal position. Then $CI(Y_{1/n}) \cong \mathrm{Cone}(F_n)$ as $\mathbb{Z}/8$-graded complexes, with the Chern–Simons value of the generator corresponding to x equal to $A(x) + \delta_n(x) + c_i$. The differential of $CI(Y_{1/n})$ is* not *assumed to be a bigon count; only its generators and levels are identified.*

Proof sketch. (i), (ii) are Propositions 1.2, 2.2 with the perturbation term; (iii) is Floer's filtration argument, which we take from [6]; (iv) is the standard continuation argument together with the first-variation formula for the perturbed functional; the integral equals $\delta_n(x)$ by (ii) and Stokes. □

What remains to be checked for a complete proof: that Floer's filtration argument identifies the connecting map with the cobordism map F_n (it does up to homotopy; exactness at the level of filtered complexes needs the chain-level identification, which is what the perturbation picture provides), and the approximation of Z_n by shearing isotopies with control of the perturbed Chern–Simons values. For the latter, Appendix A shows: Sivek–Zentner's approximation is C^r for every fixed r (not C^∞), a diagonal argument (round the corners of Z_n, then approximate in C^1) makes the actions converge, but the perturbations themselves do not converge, so uniform compactness for the perturbed complexes along the sequence is open. The cleanest route avoids the limit: fix one smoothing Z_n^ϵ, use the exact formula for its actions, and let δ_n absorb the $O(\epsilon)$ difference.

6. All slopes: Farey triads

Theorem 4.2 and Theorems 5.3–5.4 were stated for the triads $(1/(n-1), 0^w, 1/n)$. Nothing in their proofs uses these slopes beyond two facts: the three lines are pairwise at distance one, and the complexes involved are irreducible-only complexes that are defined. Under (H1) the second fact holds for every slope.

6.1. **Reducibles for every slope.** Write $\varepsilon \in \{0,1\}$ for the two SO(3) bundles on $Y_{p/q}$ when p is even ($\varepsilon = 1$ nontrivial), and $\varepsilon = 0$ when p is odd. Flat connections on $(Y_{p/q}, \varepsilon)$ are the $\rho \in R(X)$ with $\rho(\mu^p\lambda^q) = (-1)^\varepsilon$, i.e. $r(\rho) \in L^\varepsilon_{p/q} := \{pa + qb \equiv \varepsilon/2\}$; for p odd the two lines are exchanged by χ and describe the same SO(3) connections (Proposition 1.1(c)).

Lemma 6.1. *Assume (H1). For every p/q and ε: (a) the reducible flat connections on $(Y_{p/q}, \varepsilon)$ are the abelian ρ with $\rho(\mu)^p = (-1)^\varepsilon$, i.e. the points $(a, 0)$ with $pa \in \varepsilon/2 + \mathbb{Z}$; (b) each is nondegenerate; (c) the irreducible flat connections form a compact set at positive distance from the reducibles, uniformly in the slope, namely $r^{-1}(L^\varepsilon_{p/q}) \subset R^{\mathrm{irr}}(X)$.*

Proof. (a) Abelian representations have $\rho(\lambda) = 1$. (b) For $\rho(\mu) = \pm 1$ (central) nondegeneracy is $H^1(Y_{p/q}; \mathbb{R}) = 0$. Otherwise $\operatorname{ad}\rho = \mathbb{R} \oplus \mathbb{C}_\chi$ with $\chi(\mu) = \rho(\mu)^2 = e^{2\pi i \cdot 2a}$, $\chi \neq 1$, and $H^1(Y_{p/q}; \mathbb{C}_\chi)$ sits in the Mayer–Vietoris sequence with $H^1(X; \mathbb{C}_\chi)$, which vanishes iff $\Delta_K(e^{4\pi i a}) \neq 0$, and $H^1(S^1 \times D^2; \mathbb{C}_\chi) = 0 = H^0(T^2; \mathbb{C}_\chi)$; so $H^1(Y_{p/q}; \operatorname{ad}\rho) = 0$ by (H1). (c) is Proposition 1.1(a): $R^{\mathrm{irr}}(X)$ is compact and disjoint from the abelian locus. □

We take as given (it is the premise of this section) that for a rational homology sphere all of whose reducible flat connections are nondegenerate, the irreducible-only instanton complex $CI(Y_{p/q}, \varepsilon)$ is defined, with the usual $\mathbb{Z}/8$ grading and Chern–Simons filtration; by Lemma 6.1(c) no perturbation near the reducibles is ever needed, and the generators are $r^{-1}(L^\varepsilon_{p/q})$ by Proposition 1.2, whose proof is slope-independent. Proposition 2.2 is already stated for arbitrary p/q.

6.2. **Farey triads and the bundle parity rule.** Let $\alpha = p/q$ and $\beta = r/s$ be Farey neighbours, $ps - qr = \pm 1$, with direction vectors $v_\alpha = (q, -p)$, $v_\beta = (s, -r)$, and let $\gamma = \alpha \oplus \beta = (p + r)/(q + s)$ be the mediant, $v_\gamma = v_\alpha + v_\beta$. Any rational slope is reached from $1/0, 0/1$ by iterated mediants (the Stern–Brocot tree). For a half-lattice point $c \in \frac{1}{2}\mathbb{Z}^2$ and a primitive direction v, the line $c + \mathbb{R}v$ is L^ε_v with $\varepsilon(c, v) = 2\det(c, v) \bmod 2$.

Lemma 6.2 (parity rule and zero flux)**.** *Let $L^{\varepsilon_\alpha}_\alpha$ and $L^{\varepsilon_\beta}_\beta$ meet at $\mathsf{p} \in \frac{1}{2}\mathbb{Z}^2$ (a corner of P). The Lagrangian surgery $Z = L_\alpha \#_{\mathsf{p}} L_\beta$ is Hamiltonian isotopic in T^2 to the line of direction v_γ through the midpoints $m_1 = \mathsf{p} + \frac{1}{2}v_\alpha$, $m_2 = \mathsf{p} + v_\alpha + \frac{1}{2}v_\beta$, which is $L^{\varepsilon_\gamma}_\gamma$ with*

$$\varepsilon_\gamma \equiv \varepsilon_\alpha + \varepsilon_\beta + 1 \pmod 2.$$

Proof. Zero flux as in Lemma 5.1, with $v_1 = v_\alpha$, $v_2 = v_\beta$ (which only used $\det(v_1, v_2) = \pm 1$). Then $2\det(m_1, v_\gamma) = 2\det(\mathsf{p}, v_\alpha) + 2\det(\mathsf{p}, v_\beta) + \det(v_\alpha, v_\beta) \equiv \varepsilon_\alpha + \varepsilon_\beta \pm 1$, using $\mathsf{p} \in L_\alpha \cap L_\beta$. □

Checks: $(1/0,\, 0^w) \mapsto \varepsilon = 0 + 1 + 1 = 0$, the line L_1 through the origin (Floer's triangle, as verified in §7); $(1/0,\, 0^0) \mapsto L^w_1$; $(1/1,\, 1/0) \mapsto L^w_2$, i.e. Y_2 with its *nontrivial* bundle, while L^0_2 arises from $(1/1^w, 1/0)$. This is the pillowcase form of the bundle bookkeeping in a surgery triad.

6.3. **The theorem for all slopes.** Fix a Farey triad (α, β, γ) with bundle classes satisfying the parity rule, and put $Q_\alpha, Q_\beta, Q_\gamma$ for the quadratic forms of Proposition 2.2 (any choice of complementary basis; the choice changes nothing modulo $\mathbb{Z}$). Define the *cone actions* on the two legs by

$$A_{\mathrm{cone}}(x) = \begin{cases} f_\Gamma(\tilde{x}) - Q_\alpha(\tilde{x}) + \kappa_\alpha, & x \in \Gamma_K \cap L_\alpha, \quad \kappa_\alpha := Q_\alpha(m_1) - Q_\gamma(m_1), \\ f_\Gamma(\tilde{x}) - Q_\beta(\tilde{x}) + \kappa_\beta, & x \in \Gamma_K \cap L_\beta, \quad \kappa_\beta := Q_\beta(m_2) - Q_\gamma(m_2), \end{cases}$$

so that the primitives on the two legs of Z agree with the primitive of L_γ at the points m_1, m_2 where Z meets L_γ. Since the Q's have integer coefficients and $m_i \in \frac{1}{2}\mathbb{Z}^2$, $\kappa_\alpha, \kappa_\beta \in \frac{1}{4}\mathbb{Z}$; for the triad $(1/(n-1), 0^w, 1/n)$ they are 0 and $\frac{n}{4}$ (§7).

Theorem 6.3 (mapping cone for an arbitrary slope)**.** *Assume (H1)–(H3), that Γ_K is transverse to $L_\alpha, L_\beta, L_\gamma$, and that Γ_K is in minimal position with respect to the triad ($|\Gamma_K \cap L_\gamma| = |\Gamma_K \cap L_\alpha| + |\Gamma_K \cap L_\beta|$; for $K \subset S^3$, where Γ_K has homology class $(0,k)$, this holds whenever Γ_K meets each of the three lines minimally, because $q_\gamma = q_\alpha + q_\beta$). Then, with F the map of the 2-handle cobordism,*

$$CI(Y_\gamma, \varepsilon_\gamma) \;\cong\; \mathrm{Cone}\big(F\colon CI(Y_\alpha, \varepsilon_\alpha) \to CI(Y_\beta, \varepsilon_\beta)\big)$$

as $\mathbb{Z}/8$-graded complexes, and the Chern–Simons value of the generator of $CI(Y_\gamma)$ corresponding to a cone generator x is

$$\mathrm{cs}(x') = A_{\mathrm{cone}}(x) + \delta(x) + c_i, \qquad \delta(x) = \int_{R(x)} \omega,$$

with $R(x)$ the region bounded by the arc of $\tilde{\Gamma}_K$ from $\tilde{x}$ to $\tilde{x}'$, the leg of Z from $\tilde{x}$ to its midpoint m, and the segment of L_γ from m to $\tilde{x}'$. Modulo the same ingredients as Theorem 5.4 (Floer's filtration argument for the perturbed complex, now on rational homology spheres with nondegenerate reducibles, and the perturbed action formula of Appendix A).

Proof sketch. Identical to Theorems 5.3 and 5.4 with Lemma 6.2 in place of Lemma 5.1. The perturbed critical set on Y_γ is $r^{-1}(L_\alpha) \sqcup r^{-1}(L_\beta)$ together with the reducibles $Z \cap \{b = 0\} = (L_\alpha \cup L_\beta) \cap \{b = 0\}$, i.e. the reducibles of Y_α and of Y_β ($p_\gamma = p_\alpha + p_\beta$ of them, consistent with H_1); by Lemma 6.1 they are nondegenerate and isolated from the irreducibles throughout the perturbation, so the irreducible-only complexes are defined along the whole family. □

Corollary 6.4 (Stern–Brocot recursion)**.** *For $K \subset S^3$ and any p/q, $CI(S^3_{p/q}(K), \varepsilon)$ is an iterated cone over the complexes $CI(S^3_0(K), w)$ and $CI(S^3_1(K))$ ($CI(S^3) = 0$ and $CI(S^3_0, 0)$ never occur, since the Stern–Brocot path from $\{1/0, 0/1\}$ can be routed through $(1/0, 0^w) \to 1$ first), with all Chern–Simons values given by the formula of Theorem 6.3 applied along the path. In particular ± 2: $L^w_2 = \mathrm{Cone}(L_1 \to L_{1/0})$ and $L^0_2 = \mathrm{Cone}(L^w_1 \to L_{1/0})$, where $L^w_1 = \chi(L_1)$; both bundles on $Y_{\pm 2}$ are covered.*

The counting version (Theorem 4.2(a)) is the surgery exact triangle for the triad, and the bound version (Theorem 4.2(b)) holds with the levels of the triad's cobordism maps; neither is exact, for the reason given at the start of §5.

7. Tests

Scripts, outputs and plots are in `numerics/`. Ground truth: the Fintushel–Stern values $\mathrm{cs} = -e^2/(4a_1a_2a_3)$ on $\Sigma(a_1, a_2, a_3)$ oriented as the link of the singularity. Orientations were fixed independently of cs: $S^3_{1/n}(\mathrm{RHT}) = -\Sigma(2, 3, 6n-1)$, $S^3_{-1/n}(\mathrm{RHT}) = +\Sigma(2, 3, 6n+1)$, and $S^3_{+1}(4_1) = +\Sigma(2, 3, 7)$. Throughout, $\mathrm{cs}_{\mathrm{KK}}(Z) = -\mathrm{cs}_{\mathrm{FS}}(Z)$: the Kirk–Klassen normalisation is the negative of Fintushel–Stern's on every manifold tested.

7.1. **Right-handed trefoil: Proposition 2.2 holds exactly.** $\pi_1 = \langle x, y \mid x^2 = y^3 \rangle$, $\mu = y^{-1}x$, $\lambda = x^2\mu^{-6}$. The irreducible curve is the segment $b = \frac{1}{2} - 6a$, $a \in [\frac{1}{12}, \frac{5}{12}]$, from the reducible point $(\frac{1}{12}, 0)$ to $(\frac{5}{12}, -2)$, and $f_\Gamma(a) = 6(a - \frac{1}{12})^2$ measured from the trivial connection along $b = 0$. On $S^3_{1/n}$ the formula gives the closed form $\mathrm{cs} = -(6m - 6n + 1)^2/(24(6n-1))$, which is $-e^2/(4P)$ in disguise.

surgery	cs from Proposition 2.2	manifold	Fintushel–Stern
$+1$	$71/120,\ 119/120$	$-\Sigma(2,3,5)$	agrees with $-\mathrm{cs}_{\mathrm{FS}}$
$+1/2$	$95, 167, 215, 263\ /264$	$-\Sigma(2,3,11)$	agrees
$+1/3$	$47, 191, 239, 263, 359, 407\ /408$	$-\Sigma(2,3,17)$	agrees
-1	$25/168,\ 121/168$	$+\Sigma(2,3,7)$	agrees
$-1/2$	$25, 121, 217, 289\ /312$	$+\Sigma(2,3,13)$	agrees
$-1/3$	$25, 73, 121, 289, 313, 385\ /456$	$+\Sigma(2,3,19)$	agrees

All 26 points agree to 10^{-12}, *provided* the line term is read as the quadratic form Q of Lemma 2.1 evaluated at the lifted endpoint (reading (I2) in `trefoil_output.txt`); evaluating $pq\,s^2$ on the projection to $L_{p/q}$ fails whenever the lifted point lies on a translate $pa + qb = k \neq 0$.

7.2. **Right-handed trefoil: the naive corner-offset identity is false.** $\Gamma \cap \{b = \frac{1}{2}\}$ consists of $a = \frac{1}{6}$ and $a = \frac{1}{3}$. No reading of the action, no sign, and no offset in $\{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$ makes $\{\mathrm{cs}(S^3_{1/2})\}$ equal to $\{\mathrm{cs}(S^3_1)\} \cup \{A(x) + \text{offset}\}$: the denominators are 264, 120 and 24. This refutes the generator-level reading of the cone that the first draft of Theorem 4.2(c) contained, and it is as expected: a filtered homotopy equivalence preserves the filtered homotopy type, not the levels of generators. A side observation: on the line $b \in \frac{1}{2} + \mathbb{Z}$ the action is only defined modulo $\frac{1}{4}$ (changing the auxiliary d in Q shifts it by $b^2 = \frac{1}{4}$), which is the pillowcase form of the quarter-integer ambiguity of instanton energies on cobordisms with $w^2 \neq 0$.

7.3. **Figure-eight knot.** With $\pi_1 = \langle x, y \mid wx = yw\rangle$, $w = x^{-1}yxy^{-1}$, Riley's parametrisation gives the irreducible circle Γ over $a \in [\frac{1}{6}, \frac{1}{3}]$, symmetric under $b \mapsto -b$, with folds of a at $(\frac{1}{6}, \frac{1}{2})$ and $(\frac{1}{3}, \frac{1}{2})$, one transverse double point at $(\frac{1}{4}, 0)$, and *no branch points* (the restriction map is an immersion; $\min|(\dot a, \dot b)| = 0.226$). The lift of one circuit translates by $(0, \mp 2)$ and $\oint 2b\,da = \frac{1}{3}$, so f_Γ does *not* descend to Γ: it is a section of the Kirk–Klassen circle bundle, and the transition term makes the corrected period -1 (Lemma 2.4). The absolute constant from the $SL(2,\mathbb{C})$ path of Andersen–Hansen is $c = -\frac{1}{6}$, and then

surgery	relative cs	with $c = -1/6$	manifold
$+1$	$53/168,\ 149/168$	$25/168,\ 121/168$	$+\Sigma(2,3,7)$
-1	$3/168,\ 75/168$	$143/168,\ 47/168$	$-\Sigma(2,3,7)$
$\pm 1/2$	$\pm\{0.171883, 0.946066, 0.764840, 0.485874\}$		hyperbolic, no ground truth

$c = -\frac{1}{6}$ is the unique constant placing both ± 1 surgeries in $\pm\Sigma(2,3,7)$; this confirms the procedure of §2 for the absolute constants. The $\pm 1/2$ values are predictions.

7.4. **Figure-eight knot: the exact cone (Theorem 5.3).** Zero flux (Lemma 5.1): the two triangles between one period of Z_n and $L_{1/n}$ have ω-areas $\pm\frac{1}{4}$, for $n = 1, 2, 3$. Minimal position: $2 = 0 + 2$, $4 = 2 + 2$, $6 = 4 + 2$. An explicit equivariant isotopy (vertex moving linearly to the midpoint of $m_1 m_2$) keeps the count constant through 400 steps; no birth–death. With A_{cone} as in Proposition 5.2, $A(x') - A_{\mathrm{cone}}(x) = \int_{R_n(x)} \omega$ holds to $2 \cdot 10^{-10}$ for all generators, $n = 1, 2, 3$:

n	$\tilde x$ (leg)	$\tilde x'$	A_{cone}	$A(x')$	$\delta_n = $ area
1	$(\frac{1}{6}, \frac{1}{2})$ (L_0^w)	$(0.205251, 0.794749)$	$\frac{1}{6} + \frac{1}{4}$	$53/168$	$-17/168$
1	$(\frac{1}{3}, -\frac{1}{2})$ (L_0^w)	$(0.315663, -0.315663)$	$\frac{2}{3} + \frac{1}{4}$	$149/168$	$-5/168$
2	$(\frac{1}{6}, \frac{1}{2})$ (L_0^w)	$(0.170918, 0.414541)$	$0.166667 + \frac{1}{2}$	0.652541	-0.014126
2	$(0.315663, -0.315663)$ (L_1)	$(0.281081, -0.140540)$	0.886905	0.931507	$+0.044602$
2	$(\frac{1}{3}, -\frac{1}{2})$ (L_0^w)	$(0.319618, -0.659809)$	$0.666667 + \frac{1}{2}$	0.112733	-0.053934
2	$(0.205251, -1.205251)$ (L_1)	$(0.225016, -1.112508)$	0.315476	0.338550	$+0.023073$

(actions relative to $f_\Gamma(\frac{1}{6},\frac{1}{2})=0$; add $c=-\frac{1}{6}$ for absolute values). For $n=1$ the cone actions $\{0,\frac{1}{2}\}+\frac{1}{4}$ plus the areas $\{-\frac{17}{168},-\frac{5}{168}\}$ give exactly $\{\frac{25}{168},\frac{121}{168}\}$, the Chern–Simons invariants of $-\Sigma(2,3,7)$; no constant shift could (the cone actions differ by $\frac{1}{2}$, the targets by $\frac{4}{7}$). The normalisation of the w-leg was discovered here: the vertex offset $(n-1)/4$ fails, the midpoint-matched offset $\frac{n}{4}$ passes. Files: `numerics/fig8_cone.py`, `fig8_cone_n1.png`.

7.5. **Figure-eight knot: Farey triads (Theorem 6.3).** All twelve bundle combinations of the triads $(1,1/0)\to 2$, $(1/2,1)\to 2/3$, $(2,1)\to 3/2$ pass: zero flux (triangles of ω-area $\pm\frac{1}{4}$), the balanced line is the one predicted by the parity rule $\varepsilon_\gamma=\varepsilon_\alpha+\varepsilon_\beta+1$, counts are additive ($2|q|$ points on slope p/q, as Γ_K has class $(0,2)$), the equivariant isotopy has no birth–death, and $A(x')=A_{\mathrm{cone}}(x)+\int_{R(x)}\omega$ holds to $3\cdot10^{-10}$. The offsets $(\kappa_\alpha,\kappa_\beta)$ found: $(\frac{1}{4},0)$ for $L_1^0,L_{1/0}^0\to L_2^w$; $(0,\frac{1}{2})$ for $L_1^0,L_{1/0}^w\to L_2^0$; $(\frac{1}{4},0)$, $(0,\frac{1}{4})$, $(\frac{1}{4},0)$, $(0,\frac{3}{4})$ for the four $2/3$ cases; $(0,\frac{1}{4})$, $(0,\frac{1}{2})$, $(\frac{1}{2},0)$, $(0,\frac{1}{4})$ for $3/2$. As the numerics agent observed, once κ is defined by matching primitives at the midpoints the action identity is Stokes' theorem; the content of the test is the parity rule, zero flux, minimal position, the bijection, and the independence of κ from the choice of corner lift (translating m by a lattice vector leaves κ unchanged mod $\mathbb{Z}$).

Flag. L_2^w and $L_{2/3}^w$ pass through the transverse double point $(\frac{1}{4},0)$ of Γ_K, which is also the image of the abelian representation with $a=\frac{1}{4}$: two irreducible generators and one reducible share a pillowcase image. The generator count is unaffected (each branch is transverse), but any bigon model must treat this coincidence; the reducible is nondegenerate by Lemma 6.1. On the torus the reducible counts add, $p_\gamma=p_\alpha+p_\beta$; in the pillowcase they do not, because of the identifications at corners. Files: `numerics/fig8_farey.py`, `fig8_farey_output.txt`.

8. Workflow

The orchestration is one coordinator and specialised agents. Each task lists the model, the reasoning effort, its inputs, and an acceptance criterion. Tasks 1–3 ran during the preparation of this document; their outputs are summarised in §7.

(1) **Numerics (Opus, high).** Explicit SU(2) representation curves for the trefoil and the figure-eight, pillowcase images, primitives f_γ, intersection points with $L_{1/n}$ ($n=\pm1,\pm2,\pm3$) and L_0^w, the periods of Lemma 2.4, and the comparison with the tabulated Chern–Simons invariants of $\Sigma(2,3,5)$, $\Sigma(2,3,7)$, $\Sigma(2,3,11)$. Acceptance: the sign convention of Proposition 2.2 is fixed, and the offset $(n-1)/4$ of Lemma 4.1 is confirmed or refuted on $S^3_{1/2}$(trefoil).

(2) **Literature precision (Opus, medium).** Exact statements of Kirk–Klassen, Klassen, Herald, Kirk–Klassen–Ruberman, Braam–Donaldson, Daemi, Nozaki–Sato–Taniguchi, Fintushel–Stern. Acceptance: every citation in this document carries a verified statement.

(3) **Adversarial referee (Opus, extra-high).** Independent check of Lemmas 2.1, 2.4, 4.1, Propositions 1.1, 1.2, and the Dehn-twist triangle. Acceptance: each claim marked correct, corrected, or given a counterexample.

(4) **Gradings (Opus, high).** Derive the exact $\mathbb{Z}/8$ splitting formula (Maslov term plus solid-torus term) from [3], and verify it on $\Sigma(2,3,5)$, $\Sigma(2,3,7)$ and $\Sigma(2,3,11)$ against the Fintushel–Stern gradings.

(5) **Absolute constants c_i (Opus, high).** Implement the $SL(2,\mathbb{C})$ path integral for the figure-eight knot and compare with the Fintushel–Stern values for $\Sigma(2,3,7)$.

(6) **Floer's perturbation argument at chain level (Opus, extra-high).** From [6], write out the two-step filtration of $CI(Y_{1/n};\pi_n)$ and prove Theorem 5.4(iii): the connecting map is F_n at chain level, with orientations over $\mathbb{Z}$.

(7) **Collar perturbations with action control (Opus, high).** Upgrade the C^0 approximation of [20, Thm. 4.5] to control of the perturbed Chern–Simons values, so that Theorem 5.4(ii) holds exactly in the limit.
(8) **Area corrections (Opus, high).** Compute $\delta_n(x)$ for the figure-eight knot, $n = 1, 2$, verify Theorem 5.3(b) numerically, and check minimal position. (Running.)
(9) **Bigons versus instantons for $(1,1)$-type examples (Opus, extra-high).** Formulate the adiabatic limit along T^2 for $Y_{1/n}$ and identify the obstructions, as in the companion notes `cfk-gauge/`.

Routing rule: any agent that finds a sign or normalisation conflict stops and reports; the coordinator updates the conventions in §2 before downstream tasks resume.

Appendix A. Appendix: collar holonomy perturbations, shears, and the perturbed action

Prepared by the collar-perturbation agent (Task 9); tags [V]/[P]/[O] as in §8.

Setup. $Y = Y_{1/n}(K) = X \cup C \cup V$, $C = [0,1]_s \times T^2$, $\{0\} \times T^2 = \partial X$, V a solid torus with meridian $\mu\lambda^n$; (μ, λ) is an oriented basis of ∂X (the opposite convention replaces λ_0, H below by $-\lambda_0, -H$). For $A = i\sigma\omega$, $\sigma = \mathrm{diag}(1,-1)$, $a = \frac{1}{2\pi}\int_\mu \omega$, $b = \frac{1}{2\pi}\int_\lambda \omega$; $cs(A) = \frac{1}{8\pi^2}\int \mathrm{tr}(A \wedge dA + \frac{2}{3}A^3)$, so $cs(i\sigma\omega) = -\frac{1}{4\pi^2}\int \omega \wedge d\omega$. On $\widetilde{P} = \mathbb{R}^2$ put $\lambda_0 = a\,db - b\,da = -2b\,da + d(ab)$, $\omega_P = d\lambda_0 = 2\,da \wedge db$; $\tilde{L} = \{a + nb = 0\}$. Perturbation datum: primitive $v = (p,q) \in H_1(T^2)$, $\ell_v = pa + qb$; H even, 1-periodic, $H(0) = 0$, $h(\mathrm{diag}(e^{2\pi i\theta}, e^{-2\pi i\theta})) = H(\theta)$; an annulus $\Sigma = [s_0, s_1] \times S^1_u \subset [0,1] \times T^2$ $(\det(u,v) = 1)$, loops $\{z\} \times S^1_v$, $\int_\Sigma \mu_\Sigma = 1$, $\Phi(A) = \int_\Sigma h(\mathrm{Hol}_z A)\,\mu_\Sigma(z)$. Nested data (v_j, H_j), $j = 1..N$ (from the X side), $cs_\pi = cs + \sum_j \Phi_j$, as in [Z, §2], [SZ, (4.4)–(4.7)]; [Z]=arXiv:1605.08530, [SZ]=arXiv:1710.01957.

(1) Shear. [V] Critical points of cs_π solve $F_A = \sum_j c\,h_j'(\mathrm{Hol}_{\iota_j} A)\mu_j$ ([Z, Prop. 1.3]; gradient of cylinder functions [KM, §3.2, Def. 3.3, (72)], KM=arXiv:0806.1053); on C every solution is reducible and boundary holonomies obey, in radians, $\alpha \mapsto \alpha + f(\beta)$, $f = g'$, conjugated by $A_k \in SL(2,\mathbb{Z})$ [Z, Prop. 2.1], [SZ, (4.8), (4.9)]; equivalently Herald's condition $\nu = f(\beta)$ on a perturbation solid torus [HHK, §7.1], HHK=arXiv:1301.0164, citing Herald, CAG 2 (1994), Lemma 61. The constants linking $CS + \Phi$ to $F = \chi'\mu$ are not tracked in [Z,SZ]. [P] In the normalization above, the abelian Euler–Lagrange equation is $d\omega = \pi H'(\ell_v)\mu_\Sigma$, whence by Stokes on Σ

$$\phi_{v,H}(x) = x + \tfrac{1}{2}H'(\ell_v(x))\,(q,-p), \qquad \iota_X\omega_P = d(H \circ \ell_v),\ \ \phi_{v,H} = \mathrm{Fl}^1_X, \tag{1}$$

i.e. the shear is the time-one Hamiltonian flow of $H \circ \ell_v$ for ω_P (the $\frac{1}{2}$ is forced by $\omega_P = 2da \wedge db$). With $\phi = \phi_N \circ \cdots \circ \phi_1$ ($\mathbb{Z}^2$- and $\pm$-equivariant, fixes the four corners, Hamiltonian): [P] gauge classes of critical points of cs_π on $Y \leftrightarrow \{[\rho] \in R(X) : i^*\rho \in L^\pi := \bar{\phi}^{-1}(L_{1/n})\}$ (proof of [SZ, Lemma 4.6] with $F \times S^1$ replaced by V; reducibles of Y are not excluded here).

(2) Approximation. [V] [SZ, Thm. 4.5, App. A]: for a smooth $\mathbb{Z}/2$-equivariant area-preserving isotopy ψ_t of T^2, any $r \geq 0$, $\epsilon > 0$, there is a piecewise-in-t isotopy through equivariant shearings ϕ_t with $\|\phi_t - \psi_t\|_{C^r(T^2)} < \epsilon$ for all t. Global C^r on T^2 for each fixed r (not C^∞; corners of P play no role upstairs); $r = 0$ is [Z, Thm. 3.3]; the error functions depend on C^{r+1} bounds of the generating vector field (App. A). Curves enter via [SZ, Lemma 4.9] (Moser), which assumes the curves avoid the orbifold points; $L_{1/n}$ passes through $(0,0)$, so a relative version (curves fixed near corners) is needed: [P], standard.

(3) Action. [P] For a perturbed-flat collar piece with input x, $\omega = 2\pi(a_u(s)d\theta_u + \ell\, d\theta_v)$, $a_u' = \frac{1}{2}H'(\ell)\cdot(\text{bump})$, so $cs(A|_{\text{piece}}) = -\frac{1}{2}\ell H'(\ell)$, $\Phi = H(\ell)$:

$$cs + \Phi = S_{v,H}(x) := H(\ell_v) - \tfrac{1}{2}\ell_v H'(\ell_v), \qquad \phi_{v,H}^*\lambda_0 - \lambda_0 = dS_{v,H}, \tag{2}$$

which is the standard formula $S = \int_0^1 (H + \lambda_0(X)) \circ \mathrm{Fl}_X^t\, dt$. Composition: $S_\phi = \sum_j S_j \circ \phi_{j-1} \circ \cdots \circ \phi_1$, $S_\phi(0) = 0$. Using locality of the CS integrand, $\delta cs_X = -\frac{1}{8\pi^2}\int_{\partial X} \mathrm{tr}(A \wedge \dot{A}) = \lambda_0(\dot{x})$ on flat normal-form paths, and $\lambda_0|_{\tilde{L}} \equiv 0$ (so cs_V is constant), for a flat path on X with lifted boundary path $\gamma \subset \mathbb{R}^2$, $\phi(\gamma(0)), \phi(\gamma(1)) \in \tilde{L}$:

$$cs_\pi(A_1) - cs_\pi(A_0) \equiv \int_\gamma \lambda_0 - \left[f_{L^\pi}\right]_{\gamma(0)}^{\gamma(1)}, \qquad f_{L^\pi}(x) := -S_\phi(x) = \int_0^x \lambda_0\big|_{\tilde{L}^\pi} \tag{3}$$

mod $\mathbb{Z}$ ($\tilde{L}^\pi = \phi^{-1}\tilde{L} \ni 0$; the integral is along $\tilde{L}^\pi$), i.e. $2\times$ the signed area between γ and $\tilde{L}^\pi$. In KK form: $cs_\pi(A_1) - cs_\pi(A_0) \equiv -2\int_\gamma b\, da + [ab + S_\phi]_{\gamma(0)}^{\gamma(1)}$; for one λ-shear ($v = (0,1)$): $[-nb^2 + H(b) - bH'(b)]$, i.e. $f_{L^\pi}^{KK} = f_L^{KK} + \big(bH'(b) - H(b)\big)$, $f_L^{KK} = nb^2$; $H = 0$ gives the unperturbed Kirk–Klassen formula (Math. Ann. 287 (1990), Thm. 4.2; not re-checked). *Correction:* the term is not $h(\rho_1) - h(\rho_0)$ alone; the collar curvature contributes $-\frac{1}{2}\ell H'(\ell)$ (λ_0-bookkeeping), equivalently $-\ell H'(\ell)$ in the KK bookkeeping. If the lifts of the endpoints lie on different $\mathbb{Z}^2$-translates of $\tilde{L}^\pi$, a gauge change on X adds the same π-independent correction as in KK [P].

(4) Limits. By (3), actions depend only on $\tilde{L}^\pi$ through 0. For arcs γ_1, γ_2 from 0, $|\int_{\gamma_1}\lambda_0 - \int_{\gamma_2}\lambda_0| \le 2\delta(\mathrm{len}\,\gamma_1 + \mathrm{len}\,\gamma_2 + |\gamma_2(1)|)$, $\delta = \|\gamma_1 - \gamma_2\|_{C^0}$. Needed [P]: perturbations π_m with $L^{\pi_m} \to Z$ uniformly as parametrized curves through 0, with $\sup_m \mathrm{len}(L^{\pi_m}) < \infty$, and intersection points $x_m \in i^*R(X) \cap L^{\pi_m} \to x$ on a fixed component of $R(X)$; C^0 alone is insufficient, C^1 is impossible at corners of Z. Z must pass through the corners of P met by L and cut P into regions of the same areas. [SZ] gives this for each fixed smoothing, not as a limit; modification [P]: round corners of Z to C_m' ($\mathrm{len}\, C_m' \le \mathrm{len}\, Z$, $C_m' \to Z$ in C^0), take ψ_m from the relative Lemma 4.9, then ϕ_m from Thm. 4.5 ($r = 1$) with $\|\phi_m^{-1} - \psi_m^{-1}\|_{C^1} < 1/m$; then $\mathrm{len}\, L^{\pi_m} \le \mathrm{len}\, C_m' + \mathrm{len}(L)/m$. [O] π_m do not converge (N_m, $\|H_j\|$ unbounded as $\|\psi_m\|_{C^2} \to \infty$), so any uniformity needing bounds on π (Floer compactness, spectral invariants, reducibles of Y) is not supplied by [SZ].

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